The Holy Grail of Holy Paladins

by Dedolox

1 Introduction

This article aims to finally clarify which stats are most effective for the healing performance of a Holy Paladin at different stages of the game. To do this, I will first define the boundary conditions, determine which healing spells should be used, and then introduce a new metric with which I create a rigorously derived point system. Finally, further considerations will be discussed, and synergy effects will be taken into account.

2 Assumptions

It is assumed, in these discussion, that the paladin has all appropriate healing talents, namely:

-Divine Intellect

-Healing Light

-Illumination

-Sanctified Light

-Holy Power

-Light's Grace

-Holy Guidance

3 Healing Spell Efficiency

Before getting to the core topic of this guide - stat evaluation - we first need to determine which healing spells are the most efficient. By efficiency, I always mean heal per mana and second (hpms).

Throughout the following sections, I will use these abbreviations:

ct = cast time

bh = bonus healing

min_h = minimum healing value of a spell (tooltip)

max_h = maximum healing value of a spell (tooltip)

spell_lvl = level at which the spell rank is learned

heal_av_nc = average heal of a non-critical cast

heal_av_c = average heal of a critical cast

heal_av = actual average heal (including crits and non-crits)

mana_av = average mana spent (including mana refund on crit)

hpm = heal per mana (hpm_av = average heal per mana)

hpms = heal per mana and second (hpms_av = average heal per mana and second)

eh = effective healing, a new metric I introduce to measure the additional healing per mana and second gained from a stat

First, we only consider Holy Light ranks 4 and higher. The table below lists tooltip healing values and mana costs.

Important:

Mana costs do not include reductions from "Libram of Absolute Truth" and healing values already include the 12% bonus from the talent “Healing Light”

Rank	Spelllevel	Manacosts	Healvalue
Rank 4	22	190	360 - 413
Rank 5	30	275	567 - 637
Rank 6	38	365	803 - 895
Rank 7	46	465	1084 - 1206
Rank 8	54	580	1424 - 1584
Rank 9	60	660	1813 - 2015
Rank 10	62	710	1985 - 2208
Rank 11	70	840	2459 - 2740

Table 1: Base Stats Holy Light

Each spell does not heal for a fixed value but for a range between min_heal and max_heal. Since all values in that range are equally likely, the expected value is simply the midpoint:

$$\frac{min_heal + max_heal}{2} \qquad (1)$$

To keep the math manageable, I ignore variance. The expected value is sufficient for our purposes.

The additional healing gained from bonus healing depends on the cast time and the level at which the spell is learned

The scaling formula is:

$$bh \cdot \frac{ct}{3.5} \cdot \frac{spell_lvl + 11}{70} \qquad (2)$$

The term (spell_lvl + 11)/70 is capped at 1.

In general, bonus healing and spell damage are normalized to a 3.5 second cast time.

Thus, the average non-crit heal is just the addition of (1) and (2) and becomes:

$$heal_av_nc = \frac{min_heal + max_heal}{2} + bh \cdot \frac{ct}{3.5} \cdot \frac{spell_lvl + 11}{70} \cdot 1.12 \qquad (3)$$

The factor 1.12 comes from "Healing Light" and only applies to Paladins. For the following calculations I assume: 2000 bonus healing and 20% crit. The talent "Holy Power" gives me an additional crit of 5% on holy spells, so we actually have 25% crit. For Holy Light we get an additional crit of 6% of "Sanctified Light". With two T6 items we get again 5% crit on Holy Light.

The goal of the calculation is to first determine the average healing per mana (hpm_av) and then divide this value by the cast time (for example, two seconds for Holy Light) at a later stage. Haste does not need to be considered here, since it has no influence on healing per mana. To do this, I first calculate the average healing per mana for a non-critical heal (heal_av_nc) using Equation (3). Afterwards, I additionally determine the average healing of a critical heal (heal_av_c).:

$$heal_av_c = heal_av_nc \cdot 1.5 \qquad (4)$$

To now obtain the actual average heal per spell for the respective rank (heal_av), I only need to calculate the weighted mean:

$$heal_av = heal_av_nc \cdot (1 - real_crit) + heal_av_c \cdot real_crit \qquad (5)$$

Real_crit is meant to reflect that talents are taken into account. With 20% crit, the spell Holy Light has a real_crit value of 31%.

The average mana consumed (mana_av) is calculated similarly, since it must be considered that a critical heal spell only costs 40% of the normal mana:

$$mana_av = mana \cdot (1 - real_crit) + mana \cdot real_crit \cdot 0.4 \qquad (6)$$

Finally, we obtain our target value, the average healing per mana per second (hpms_av) that we actually achieve per cast heal spell, by calculating:

$$hpms_av = \frac{heal_av}{mana_av \cdot ct} \qquad (7)$$

This results in the following values for Holy Light:

Rank	hpms_av
4	4.26
5	3.97
6	3.83
7	3.74
8	3.66
9	3.78
10	3.69
11	3.56

Table 2: Holy Light No Libram

Rank	hpms_av
4	4.52
5	4.21
6	4.06
7	3.96
8	3.88
9	4.01
10	3.92
11	3.76

Table 3: Holy Light 2T6 No Libram

Rank	hpms_av
4	4.38
5	4.08
6	3.92
7	3.82
8	3.74
9	3.85
10	3.77
11	3.61

Table 4: Holy Light LotL

Rank	hpms_av
4	4.65
5	4.33
6	4.16
7	4.06
8	3.97
9	4.09
10	3.99
11	3.83

Table 5: Holy Light 2T6 LotL

Rank	hpms_av
4	5.19
5	4.53
6	4.22
7	4.03
8	3.89
9	3.98
10	3.88
11	3.70

Table 6: Holy Light LoAT

Rank	hpms_av
4	5.51
5	4.81
6	4.48
7	4.28
8	4.12
9	4.23
10	4.12
11	3.92

Table 7: Holy Light 2T6 LoAT

This only confirms what was already known before: Holy Light Rank 9 is the most efficient healing spell, and the "Libram of Absolute Truth" is the most efficient libram. Particularly noteworthy is the strong increase in hpms_av from the additional 5% crit provided by T6.

4 Stats

4.1 Criticism of the existing point system

In this section, I would like to take a very detailed look at the most important stats for the Holy Paladin and then develop this into a largely consistent point system. I was inspired in this by the excellent Healadin thread on Elitist Jerks from Zurm. However, I do not consider his point system to be logically derived.

What bothers me about his point system is actually quite simple: he formulates a postulate without justifying it. The starting point of his point system is the following assumption:

$$18bh = 8int = 8crit = 3mp5 \qquad (8)$$

That all of these values are completely equal for the Holy Paladin is his assumption for the further calculations, but unfortunately no explanation is given as to why this should be the case. From this equation, for example, it follows that if one sets $1bh = 1pkt$:

$$18bh = 18pkt = 8int \Leftrightarrow 1int = 2.25pkt \qquad (9)$$

If one accepts the postulate, the point system itself is not objectionable. However, Zurm subsequently revises his own premise by additionally attributing the extra crit and bonus healing to the Intellect stat. In doing so, he contradicts his original assumption and thereby undermines the entire point system itself. However, I would still like to investigate whether this postulated equality is actually correct and, hopefully, develop a more precise point system.

4.2 Effective Healing & Point-System

My idea is that I can only meaningfully compare values that share the same unit. For this purpose, I introduce effective healing (eh). This value is intended to indicate the impact that one point of a stat has on my hpms value. The approach is actually quite simple: I have already calculated the hpms_av values for all Holy Light ranks, and now I change the initial conditions one at a time. For 1 bonus healing, for example, I use 2001 bonus healing, while everything else remains the same.

I then calculate hpms_av again using exactly the same formula. Now I simply compute the difference between the old and the new hpms_av and obtain my eh value for the corresponding rank of the healing spell.

An example:

$$eh_rank9 = hpms_av_rank9(bh = 2001) - hpms_av_rank9(bh = 2000) \qquad (10)$$

Finally, however, I still have to take into account that I use two ranks of Holy Light, namely Rank 9 and Rank 11. For this calculation, I will first assume that I cast Rank 9 in 80% of cases and Rank 11 in 20% of cases (I will also perform the calculation for other distributions). In the end, I therefore have to form the weighted mean again:

$$eh = eh_rank9 \cdot 0.8 + eh_rank11 \cdot 0.2 \qquad (11)$$

And thus I obtain a value that indicates exactly how much effective healing 1 point of bonus healing provides for an average heal.

It works the same way for crit. Instead of assuming 2000 bonus healing and 20% crit, I use 2000 bonus healing and 20.04529% crit (22.08 crit rating grants 1% crit).

For mp5, I keep 2000 bonus healing and 20% crit and instead modify the mana used. During the cast time of Holy Light, 1 mp5 restores exactly $\frac{2}{5}$ mana. I simply subtract this value from the mana cost of my healing spells and then repeat the procedure.

To calculate the eh from haste, I first need to clarify how haste is applied. The formula for the new cast time (ct) is:

$$ct_new = \frac{ct_old}{1 + \frac{x}{1577}} \qquad (12)$$

Here, x represents the haste rating. 15.77 haste rating provides 1% spell cast time reduction.

The eh value for 1 haste is therefore calculated as follows:

$$\left(1 + \frac{1}{1577}\right) \cdot hpms_av - hpms_av = \frac{1}{1577} \cdot hpms_av \qquad (13)$$

Whereas hpms_av is again determined by the weighted average:

$$hpms_av = hpms_av_rank9 \cdot 0.8 + hpms_av_rank11 \cdot 0.2 \qquad (14)$$

We recall that the hpms value is determined simply by dividing the hpm value by the cast time. In this case, that would be 2. For the eh value, which takes haste into account, however, I must divide the difference in the hpm value by $\frac{2}{1+\frac{x}{1577}}$. Since the hpms value has already been divided by 2, the denominator remains as a multiplication. Note that although haste may at first appear not to affect cast time linearly (more on that later), it does affect hpms, and therefore eh, in a linear way.

For these four values (bh, crit, haste, mp5), an exact eh value can be determined in this way, which makes them easy to compare and ultimately allows them to be converted cleanly into a point system. Intelligence behaves somewhat differently. First, due to the talent Divine Intellect, 1 Int gives 1.1 Int directly. Together with the talent Holy Guidance, 1 Int therefore provides 0.385 bonus healing, 0.01375% crit, and 16.5 mana. Converting mana into eh is not straightforward - more on that later. So first, I calculate the eh value for 1 Int by considering only the bonus healing and the crit, and will return to the mana afterward. I therefore use 2000.385 bh and 20.01375% crit and determine the eh value as before.

I obtain the following values:

1 int	$9.1727 \cdot 10^{-4}$
1 int BoK	$10.0901 \cdot 10^{-4}$
1 haste	$23.6604 \cdot 10^{-4}$
1 crit	$19.7826 \cdot 10^{-4}$
1 bh	$8.3210 \cdot 10^{-4}$
1 mp5	$21.7050 \cdot 10^{-4}$

Table 8: effective healing values

BoK here simply means that Blessing of Kings is taken into account.

The problem with mana is that while it can be easily converted into healing, it cannot easily be converted into eh, since it does not act continuously but rather in a binary way. Mana that is actually used is extremely effective, whereas unused mana provides no benefit at all. Therefore, everything that follows must be considered an approximation, without any claim of being exact. I will first begin by neglecting the time factor and reducing my analysis to hpm rather than hpms.

I will now determine how much healing these 16.5 mana provide in total. To do this, I simply calculate the hpm_av values for Holy Light Rank 9 and Rank 11 and once again form the weighted mean:

$$hpm_av = hpm_av_rank9 \cdot 0.8 + hpm_av_rank11 \cdot 0.2 \qquad (15)$$

I now only need to multiply this value by the 16.5 mana to obtain the amount of healing that I achieve on average with 16.5 mana in a raid situation. At this point, I should emphasize once again that this is only a theoretical approximation, since 16.5 mana is of course not enough to cast a heal, and this must be kept in mind due to the binary nature of how mana functions, as mentioned earlier. The value amounts to 123.1613 healing.

To now compare the other stats with Intelligence and determine which gems and which stat are best for the Holy Paladin, I combine all hpm values of the stats (except haste, since the time dependency cannot be eliminated there) into a function Healing(Mana) and plot them.

For this, I multiply the hpm values of the individual stats by the corresponding gem values, i.e., bonus healing times 22 for epic or times 18 for blue, crit times 10 for epic or times 8 for blue, and so on.

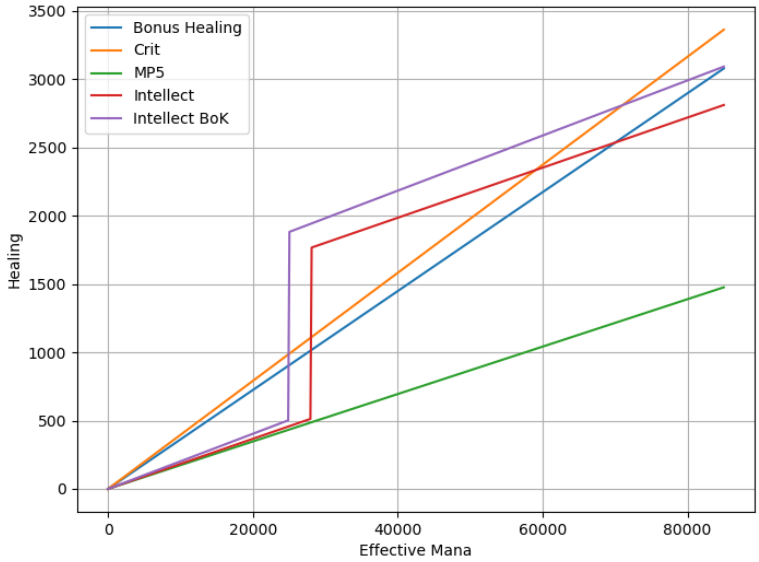

Figure 1: epig gems

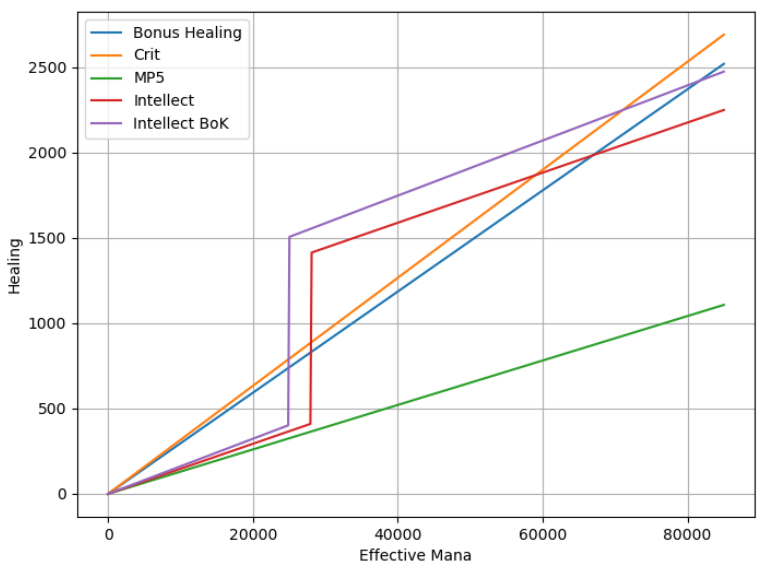

Figure 2: blue gems

The fact that I placed the spike in the Intelligence graphs, which represent healing from the 165 or 181.5 additional mana, at 25k and 28k mana is arbitrary. My aim is simply to show that Intelligence only becomes effective once a certain amount of effective mana has been consumed. Before that

point, Intelligence as a stat is simply worse than others, but as soon as you reach the mana range that you wouldn't have without the extra Intelligence, the effectiveness of Intelligence spikes and surpasses all other stats. If additional mana is not needed, the spike does not occur, and Intellect is an ineffective stat. The shift of the two Int graphs relative to each other is only for better visibility.

It can also be clearly seen that there is no significant difference in value between epic gems and blue gems.

As you can see, Zurm's postulate is not tenable, and therefore the point system based on it is also fundamentally flawed.

Finally, I would now like to derive a point system based on this. I set $1bh = 1pt$, which implies $1pt = 8.321010^{-4}$eh, and derive all other point values from this. To do this, I simply divide all eh values by the eh value of bh and obtain the following point system (here the time factor is again taken into account):

1 int	1.102
1 int BoK	1.213
1 haste	2.843
1 crit	2.377
1 bh	1
1 mp5	2.608

Table 9: point system

It is important to note that for Intelligence, mana was not taken into account. For all other stats, this is a largely precise and cleanly derived point system. As long as you have enough mana for the content you are currently doing and basically never run into mana issues, this is a complete point system, because the point value of Intelligence is only increased when the additional mana is actually needed.

For completeness, however, I also want to include the case where mana is required. Here, I have to work with averages of hypothetical scenarios, which is why the eh value for Intellect can never be as precise as for the other stats. I believe, however, that in the end a useful point system will still emerge that reasonably ranks Intellect.

My idea is to take the eh_total value, which can be calculated exactly as in formula (15) as a weighted mean, and then normalize it with the following factor:

$$eh_mana = hpms_av \cdot \frac{16.5}{\text{eff}_mana} \qquad (16)$$

Here, eff_mana stands for the effective mana available during the fight.

I calculate the scenarios with the following stats: 14,500 base mana buffed, 3,480 mana from Mana Tide Totem, 91 mp5, +49 mp5 (Improved Bow), +62.5 mp5 from Shaman Totem, 1 rotation of Mana Potion and Demonic Rune = 3,600 mana. With these values, I calculate the effective mana during a 2-minute fight and during an 8-minute fight and then take the average. I assume that the mana potion + Demonic Rune rotation is started after 1 minute into the fight and then used on cooldown. This results in an eh value of $18.4700 \cdot 10^{-4}$ for 1 Int.

Scaled up to the eh value for 1 Int from bonus healing and crit, this yields the following final point system:

1 int $\bar{m}$	1.102
1 int m	3.321
1 int $\bar{m}$ BoK	1.213
1 int m BoK	3.641
1 haste	2.843
1 crit	2.377
1 bh	1
1 mp5	2.608

Table 10: point system complete

Here, m stands for "with mana considered" and $\bar{m}$ for "without mana considered."

4.3 From Point-System to Algorithm

One might think I'm done now. I had derived a point system based on my base stats (2000 bh and 20% crit) in a clean, mathematical way. But there's a problem: I assumed fixed base stats. My system is static. It ignores synergy effects, it works without mp5 and haste as base stats, and it assigns each stat a fixed point value. When I wrote code that outputs a point system for different base stats directly using my formulas, it quickly became clear: this isn't enough. The best stats always depend on the stats you already have. When mp5 is taken into account, crit is stronger than haste at almost all times — unlike what is shown in the table. At the very beginning of the game, however, bonus healing is the strongest stat.

The importance of synergy effects can also be seen in the following example. As I showed above, haste affects hpms and thus the effective healing (eh) value in a linear way. This means that each point of haste adds the same additional hpms, no matter how much haste you already have. One might therefore think that the point value of haste should remain constant. But this is completely wrong: haste significantly boosts bonus healing and crit, and since the point system is based on comparing stats, haste becomes less valuable the more you have - simply because bonus healing and crit become stronger.

The only way to correctly evaluate all stats - and thus all items - is to take all of my formulas and build an algorithm that automatically accounts for these synergy effects. My program takes the base stats as input and then searches through the selected items to find the exact combination that produces the maximum hpms value from formula (14).

However, mana_av must still be adjusted so that MP5 is taken into account. The following applies:

$$mana_mp5 = \frac{ct}{5} \cdot mp5 \qquad (17)$$

This is the amount of mana that is restored during a spell. This value is then subtracted from the mana cost, giving us the effective mana per heal:

$$mana_eff = mana - mana_mp5 \qquad (18)$$

This leads to:

$$mana_av = mana_eff \cdot (1 - real_crit) + (mana_eff - mana \cdot 0.6) \cdot real_crit \tag{19}$$

To take overheal into account I just adjust the formula (5):

$$heal_av = heal_av_nc \cdot (1 - real_crit) \cdot (1 - overh_normal) + heal_av_c \cdot real_crit \cdot (1 - overh_crit) \tag{20}$$

4.4 How the Algorithm works

My algorithm first queries the relevant information: which items from which content you want to compare, which gems you want to use (epic or blue), and whether haste gems should be included. Additionally, the libram you are using is taken into account, as well as whether enchants should be considered. And of course, it asks whether mana is needed. If "Need Mana" is selected, the mana provided by intellect is taken into account. This should not be treated lightly, because, as mentioned, intellect becomes a very inefficient stat if the mana is not used. Therefore, one should aim to get by without it as quickly as possible.

Finally, the base stats and the share of Holy Light rank 9 are entered. Here, the following is important: if you only want to compare a headpiece, you remove your current head item and enter your base stats. If you want to optimize head + shoulders + pants at the same time, you must remove all of them first and then enter the remaining base stats otherwise you optimize for multiple head/shoulder/pants items. To optimize an entire item set, all items must be removed from your character to get the correct base stats.

My algorithm does not optimize for trinkets, so these should always be kept equipped. Alternatively, if you want to go through BiS item builds, you need to enter the stats of the BiS trinkets manually. The reason I don't include them is that extra effects, such as procs or active abilities, cannot be accurately incorporated into the value calculation.

Once everything has been entered and the items have been selected, the following happens: the algorithm first determines the two best individual items + enchants for each slot, ignoring synergy effects. The computer simply cannot test all, lets say 5^{14} or even 20^{14} combinations without crashing. After selecting two items and their corresponding enchants per slot, it forms the combination that maximizes hpms through trial and error. This gives us a preliminary best item build.

Next, it fixes all items except one - say, the headpiece - and iterates through all head items and echants with the other slots fixed, searching for the combination that maximizes hpms. It does this for every item slot. Once finished, a new best item build is generated (if any changes occurred). This process repeats until an iteration produces no further changes. There are at least three iterations.

It is important to know that sockets and meta sockets are already taken into account in this calculation. For the meta socket slot, the meta gem is applied artificially, and for normal socket slots, a +10/+8 crit gem is applied artificially. I do this for two reasons: 1) It is unfortunately not possible to optimize items and gems simultaneously, because that would create too many combinations and risk crashing the computer, and 2) I choose crit gems because crit is the most effective stat to gem for at almost all points in the game (provided you don't need mana).

If you click "Need Mana" +10/+8 int gem is applied.

Once the best item build has been found, my algorithm optimizes the sockets by iterating through all possible combinations.

After items and gems have been optimized, consumables such as food, flasks, elixirs, and oils are optimized based on the total resulting stats.

If the HPS mode is selected, the exact same process is applied. The only difference is that optimization is performed for HPS instead.

Intellect is calculated in the algorithm together with Blessing of Kings. However, the final stats are displayed without BoK for better comparability.

5 Stat Conclusion

What emerges from the calculations is that Intelligence as a stat is only valuable as long as it is actually used. That is: beyond a certain mana pool, which depends on your current gear in the raided content, Intelligence becomes ineffective and drops sharply. Up to that point, it is the absolute best stat. The mana pool should be balanced so that in important boss fights, you fall below 10-12% mana during or at the end of the fight; only then is Intelligence truly effective. Crucially, you should never have to make compromises in healing. If you reach 10% mana during or at the end of a fight but have to cast FoL in the meantime to avoid going OOM, then more Intellect is needed.

Especially at the beginning (pre-raid / T4), you should therefore socket exclusively for Intellect and generally avoid items (e.g., necklaces or rings) that have no Intellect. At this stage, it is even better to enchant a weapon with +30 Int instead of +81 bonus healing, and even 10 Int on the hands is stronger than 35 bonus healing if mana is scarce.

From T5 content onward, and especially after obtaining the Libram of Absolute Truth, you should avoid socketing Intellect as much as possible and instead prioritize more effective stats.

In general, crit remains the best stat for the Holy Paladin throughout almost the entire game. The reason haste appears significantly stronger in the point system is simply because the calculation was done without MP5. This once again illustrates how dynamic the entire system is.

Another "discovery" I made during this analysis is that Haste affects not only the hpms value linearly, but also the cast time itself. Theoretically, this is not entirely correct, because the function for calculating haste has the form: $ct_new = \frac{ct_old}{1+x}$ and it is trivial that this function is not linear. However, if you set x= haste, the function becomes: $ct_new = \frac{ct_old}{1+\frac{x}{1577}}$ - this means that we are only interested in a very small, positive range near zero, where it can be approximated linearly without significant loss of accuracy. Therefore, in practice, we can say: in WoW TBC, Haste affects hpms and cast time linearly. This means that the value of Haste does not diminish in WoW TBC, no matter how much you already have. But this is just for looking at haste in an isolated way. As I stated earlier haste gets worse in comparison to bonusheal and crit if you stack a lot of it because it makes bonusheal and crit so much better.

Another discovery that surprised me is that mp5 becomes more effective the more of it you have. This is because mp5 enters the hpms formula as follows: $\frac{A}{B-k\cdot mp5}$.

6 Stat Discussion and Limitations

As I noted earlier, it is not possible to determine an exact hpms and thus eh value for the additional mana from Intellect. Nevertheless, I believe that the final point system accurately reflects the value of Intellect, and the basic ranking order is reasonable.

What is not considered in my calculations is overheal. Here, the stats bh, crit, and Int would likely drop somewhat relative to Haste and mp5. After a few quick calculations, I cautiously assume that these corrections would not change the final ranking order. However, this would mean that Haste would become stronger relative to the other stats. (Perhaps I will do a detailed overheal calculation if I find the time.)